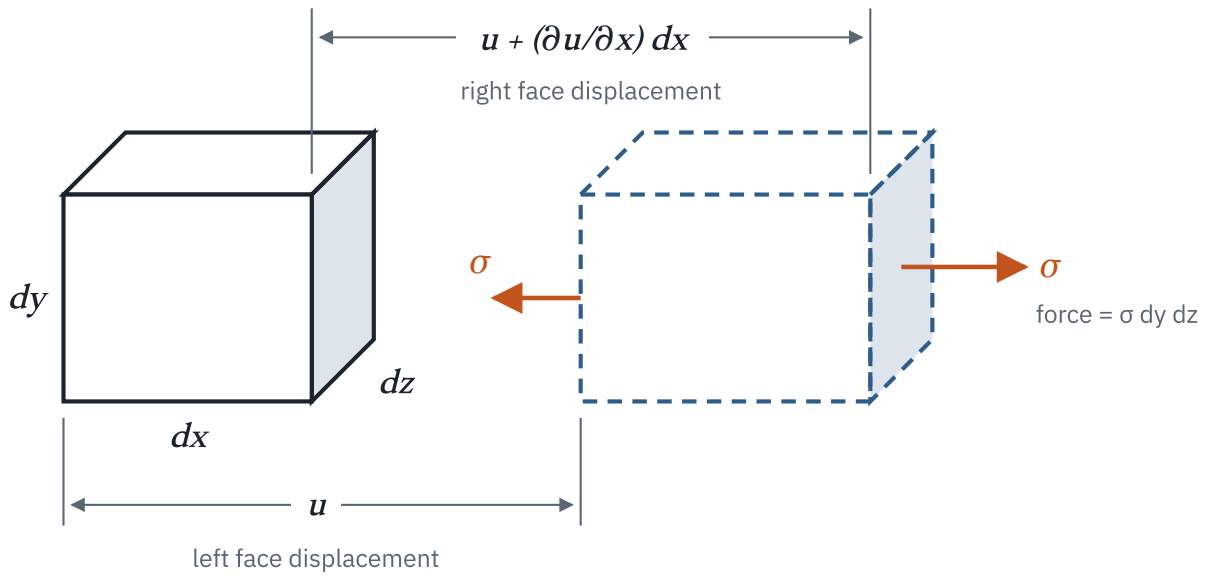
ENERGY METHODS · EQ. 6.1.1

Strain energy in a uniaxial element

A small cube is pulled by a uniaxial stress. The work done on its two faces is stored as strain energy. Equation 6.1.1 counts that work.

$$dU = \int_0^{\sigma_x} \sigma \, d\left(u + \frac{\partial u}{\partial x} dx\right) dy\, dz \; - \int_0^{\sigma_x} \sigma \, du\, dy\, dz \; = \int_0^{\sigma_x} \sigma \, d\left(\frac{\partial u}{\partial x}\right) dx\, dy\, dz \qquad (6.1.1)$$



After Fig. 6-1. Solid: the element before loading. Dashed: after loading. The cube both moves (by *u*) and stretches (by *∂u/∂x dx*).

The setup

The element has sides dx, dy, dz. The stress is raised slowly from 0 to its final value σ_x. Slow loading means no kinetic energy and no heat loss, so all the work goes into stored energy. The left face moves by u. The right face moves a little more, by $u + \frac{\partial u}{\partial x} dx$, because the material between the faces stretches. That extra term is a first-order Taylor expansion of u across the distance dx.

Term by term

1 **Work on the right face**

$$\int_0^{\sigma_x} \sigma \, d\left(u + \frac{\partial u}{\partial x} dx\right) dy \, dz$$

- The force on the face is $\sigma \, dy \, dz$, stress times area.
- It moves through a small increment of the right face's displacement.
- Force and motion both point in $+x$, so this work is **positive**.

2 **Work on the left face**

$$-\int_0^{\sigma_x} \sigma \, du \, dy \, dz$$

- The force on the left face points in $-x$, but the face moves in $+x$ by u.
- Force opposes motion, so this work is **negative**.

3 **Net work equals stored energy**

$$dU = \int_0^{\sigma_x} \sigma \, d\left(\frac{\partial u}{\partial x}\right) dx \, dy \, dz$$

- The u parts cancel. That is the rigid-body translation, which does no net work on the cube.
- Only the stretching part, $\frac{\partial u}{\partial x} dx$, stores energy.

What it means

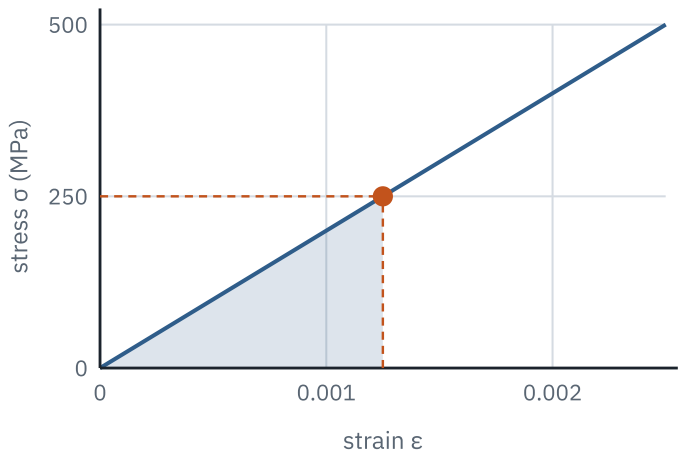
Since $\frac{\partial u}{\partial x} = \varepsilon_x$ and $dx\,dy\,dz = dV$:

$$dU = \left(\int_0^{\varepsilon_x} \sigma\, d\varepsilon\right) dV \quad \Longrightarrow \quad U_0 = \frac{dU}{dV} = \int_0^{\varepsilon_x} \sigma\, d\varepsilon$$

> U_0 is the **strain energy density**, the area under the stress–strain curve. For a linear elastic material, $\sigma = E\varepsilon$, so $U_0 = \frac{1}{2}\sigma_x\varepsilon_x = \frac{\sigma_x^2}{2E}$. The $\frac{1}{2}$ appears because the stress builds up from zero while the material deforms.

See the area

A steel element loaded to 250 MPa. The shaded triangle is U_0.



Applied stress

σ_x	250 MPa
$\varepsilon_x = \sigma_x/E$	0.00125
$U_0 = \frac{1}{2}\sigma_x\varepsilon_x$	156 kJ/m³

E = 200 GPa (steel). Doubling the stress quadruples the stored energy, because U_0 grows with σ_x^2.

Notation notes

- The dU on the left is the **energy in the small volume dV**. It is not an increment in time or load.
- The limits $0 \to \sigma_x$ describe the loading path. The integral really runs over the displacement increments $d(\cdot)$, with σ the current stress at each step. Writing the limits in strain, $0 \to \varepsilon_x$, is cleaner.
- The result $U_0 = \int \sigma\, d\varepsilon$ holds for any elastic material. The $\frac{1}{2}\sigma\varepsilon$ form needs linearity.